

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS - POWER BI REPORT

ABSTRACT:

To make a retail company understand customer purchasing behaviour to increase revenue, improve customer retention, and optimize marketing strategies.

1. DATASET TAKEN FOR ANALYSIS:

Dataset is borrowed from Internet site Kaggle. Dataset contained information focusing on analysing a dataset of Customer Shopping Behaviour to identify and understand what customers buy, how often do they buy, how much do they spend, whether they respond to discounts using Power BI visualisation tools.

2. DATASET TAKEN FOR ANALYSIS:

Dataset Name: customer_shopping_behavior.xlsx

Total Records: 5050

Data Source Type: Excel Spreadsheet

Primary Focus: Analysis of Customer Shopping Behaviour by Season, Category, Location, Shipping Type, Age, Colour, Discount Applied and Item Purchased.

3. COLUMNS IN DATASET:

- ❖ Customer ID – Unique ID for Customer
- ❖ Age – Age of Customer
- ❖ Gender – Gender Classification
- ❖ Item Purchased – Products Purchased (T- Shirt, Shorts, etc.,)
- ❖ Category – Category of Products (Clothing, Accessories etc.,)
- ❖ Purchase Amount (USD) - Cost of Products Purchased
- ❖ Location – Place of Purchase (California, Alabama, etc.,)
- ❖ Size – Size of wearing material (Small, Medium, etc.,)
- ❖ Color - Color of the Product Purchased (White, Black, etc.,)
- ❖ Season – Season of Purchase (Spring, Winter, etc.,)
- ❖ Review Rating – Points of review about particular product.
- ❖ Subscription Status – Subscription of Customers (Yes / No)
- ❖ Shipping Type – Selling Mode (Store Pickup, Shipping, etc.,)
- ❖ Discount Applied – Status of Discount on Product (Yes / No)
- ❖ Previous Purchases – Earlier purchasing count of the Customer
- ❖ Payment Method – Pay Mode for purchase (Cash, Debi Card, etc.,)
- ❖ Frequency of Purchases – Purchasing routine (Daily, Weekly, etc.,)

4. MEASURES CREATED IN ANALYSIS REPORT:

- **Average Age Group of Customers:**
AVG AGE = AVERAGE(customer_shopping_behavior[Age])
- **Maximum points of Review Rating:**
Max Review = MAX(customer_shopping_behavior[Review Rating])
- **Maximum points of Review Rating:**
MEASURE = COUNTROWS('customer_shopping_behavior')

5. ANALYSIS RESULTS:

- **Overall trend analysis:**
In these criteria, we have shown all the insights and factors for understanding the Customer Shopping Behaviour.
- **Average Age Group of Customers:**
The Overall average age group of customers is 44.15 years.
- **Maximum Points of Review Rating:**
The Overall maximum points of review rating are 5 points.
- **Products and Category Count:**
The Group of Products and Categories along with its count have been found through Countrows() Measure
- **Specific Classification**
Seasonal Response from Customers on Shopping and Trend of Payment Mode – Cash has been classified.

6. SAMPLE DATA TABLE – FIRST 10 PURCHASES:

Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color
2701	22	Female	T-shirt	Clothing	68	California	XL	Olive
521	51	Male	Sunglasses	Accessories	84	South Carolina	M	White
3157	18	Female	Shirt	Clothing	50	Montana	M	Black
1687	22	Male	Gloves	Accessories	75	Illinois	L	Red
2929	40	Female	Jewellery	Accessories	80	Alabama	L	Yellow
3583	23	Female	Shorts	Clothing	41	California	S	Beige
1135	62	Male	T-shirt	Clothing	81	Wisconsin	M	Brown
1411	61	Male	Hat	Accessories	93	Tennessee	M	Purple
113	37	Male	Gloves	Accessories	65	Florida	M	Red
166	42	Male	Handbag	Accessories	60	Kentucky	S	Yellow

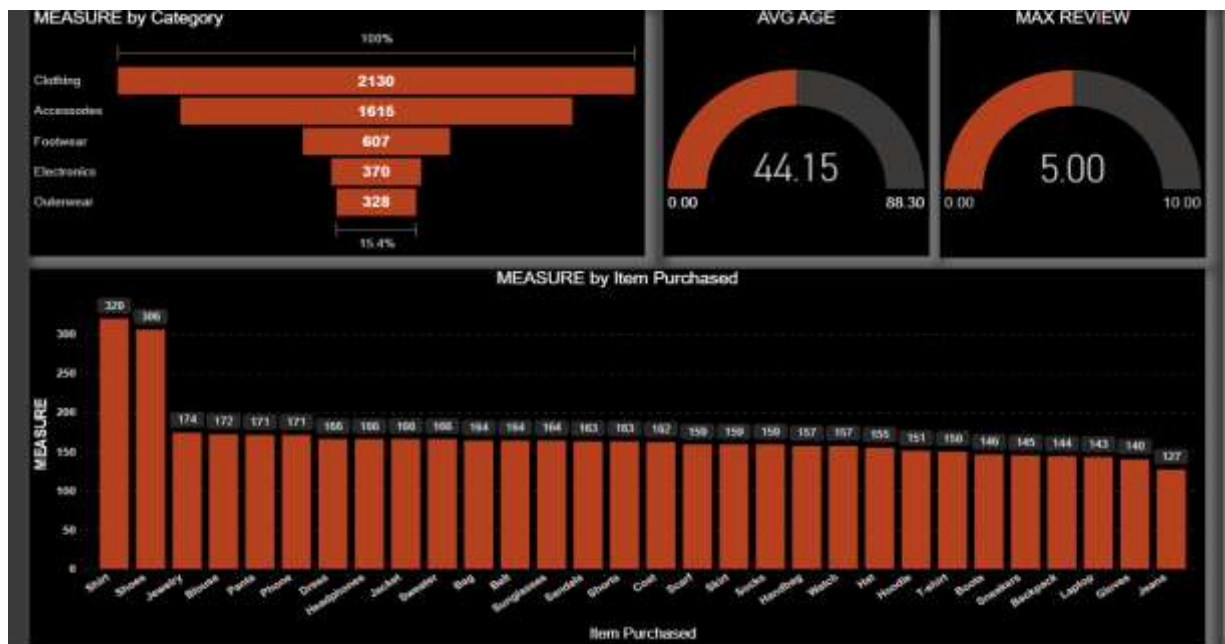
Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method	Frequency of Purchases
Winter	3.2	No	Standard	No	36	Cash	Weekly
Spring	3.9	Yes	Free Shipping	Yes	20	Debit Card	Quarterly
Winter	3.1	No	2-Day Shipping	No	18	Cash	Monthly
Fall	4.2	No	Store Pickup	No	25	Cash	Annually
Spring	3.6	No	Store Pickup	No	17	Credit Card	Weekly
Summer	3.4	No	2-Day Shipping	No	15	PayPal	Annually
Summer		No	Express	Yes	2	Venmo	Every 3 Months
Summer	4	No	Store Pickup	Yes	18	Debit Card	Annually
Spring	4.3	Yes	Free Shipping	Yes	28	PayPal	Bi-Weekly
Winter	2.8	Yes	Store Pickup	Yes	7	Cash	Monthly

7. ANALYSIS:

I. What customers buy, how often do they buy, how much do they spend, whether they respond to discounts using Power BI visualisation tools.



II. INSIGHTS OF MEASURES:



III. Seasonal Response from Customers on Shopping:



Autumn



Summer

IV. Trend of Payment Mode - Cash:



CONCLUSION:

Hence, through the analysis of Customer Shopping Behaviour it has been understood what customers buy, how often do they buy, how much do they spend, whether they respond to discounts using Power BI visualisation tools.

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Tools Used: Excel & Power BI